Annotated Bibliography

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[7] The article provides a summary of techniques presently used in sustainable development within the scientific community. The study begins by discussing the importance of sustainable development, the frequency of various techniques used to analyze sustainable development, and important issues in sustainable transportation, such as logistics and environmental issues. It then dives into the techniques that are currently being used to analyze sustainable development, which includes simulation, optimization, machine learning, and fuzzy set analysis. The Optimization section covered the history of optimization used in transportation and the variety of applications for optimization, including in the difference between urban transportation systems and in global/regional transportation systems. Applications in simulation focus on the growing integration of simulation and optimization models and on usages in logistics, traffic congestion, and on the usage of demand-based transportation systems (ride sharing applications). The machine learning section primarily summarizes a paper that uses a random forest model to understand patterns in a bike sharing transportation system, as well as touches on a variety of techniques (random forest, logistic regression, neural networks, support vector machines, decision trees, heuristics, multi-agent systems, and multi-variate analysis) for a variety of analytical, traffic-related outcomes. The section on fuzzy sets takes note of the wide range of uses that recent projects have undertaken utilizing fuzzy sets. The paper notes that sustainability factors into many of the research projects, where businesses seek sustainability to increase innovation and enhance competitiveness. The paper concludes by suggesting that more of the investigated technologies be applied in tandem with one another to increase effectiveness. The section on machine learning will be the most relevant to the IS, but the other sections provide relevant background information about other advances in the field as well as ways that these tools are used in the business world.

[8] The authors utilized a gradient-boosting machine learning algorithm in order to study traffic speeds in developing cities. By reading in data regarding road properties, day and time, and land use in the city of Cauayan in the Phillipines, the model predicted vehicle speeds in specific times and locations–generally finding that traffic speeds decreased near large urban areas. Importantly, the study notes that most similar studies into ”smart city” development occur in urban settings, while this study provides information to policymakers from developing areas. The study provides a pertinent example of a machine learning algorithm being implemented to analyze ”smart city” data to analyze urban growth in developing cities and enhance urban safety.

[3] This chapter provides a brief overview of the various techniques that combine GIS

data and machine learning currently employed in the civil engineering field. While it lists the name of the technique and a very brief paragraph about its recent advances, its upsides, and its downsides, the paper does not explain what the technique is or when it would be most beneficial to use it. There were also typos and incomplete sentences present in the writing. Neural networks, random forest and gradient boosting, support vector machines, decision trees, bayesian networks, gaussian processes, and ensemble learning all receive paragraph-long summaries. The paper then describes applications of machine learning to GIS, parimarily in the analysis of environmental hazards (flood prediction, groundwater modeling, erosion, landslip succeptibility assessment, and LULCC) but also in areas such as public health, maintaining biodiversity, or in analyzing infrastructure costs. The paper concludes with an assessment that the combined use of GIS data and machine learning methods have the potential to benefit a large variety of fields. Although the paper encompasses a breadth of information about machine learning and GIS, it is held back by its brevity and will be more useful to me as a list of keywords to research in-depth elsewhere.

[10] The article uses Random Forest, K Nearest Neighbors, XGBoost, and Artificial Neural Network to predict traffic congestion in Casablanca, Mexico, specifically the formation of traffic jams. The authors highlight the necessity of focusing on traffic jams in a quickly urbanizing city such as Casablanca. The models are trained on a mixture of historical accident data, Waze accident pixels, and GIS data to include a spatial element (road mapping, population, land usage data). Random Forest performed the best when comparing the training set data to the testing set data with 96% accuracy. K Nearest Neighbors and XGBoost were close behind with 96% and 92% respectively. In contrast, the ANN model had an accuracy of 64%. The results of traffic hotspots were then mapped onto a visual displaying high-probability traffic jam spots. The study concludes by mentioning some potential biases in the training data, such as using only Waze data, and the potential for models to perform differently under cases with more rows. The study's acknowledgement of its limitations enhances its credibility.

[1] As opposed to other studies, this study focuses on using XAI (eXplainable AI) methods to explain the "black box" machine learning models' results. The article pays special attention to the background in the application of "black box" models: their advantages over traditional statistical methods in capturing non-linearity, while the simultaneous difficulty of explaining the way it works to stakeholders and managers makes it difficult for these bodies to use. Permutation feature importance (PFI), and feature importance ranking measure (FIRM) algorithms, as well as SHAP and LIME analysis, were used in conjunction with RF, XGBoost, SVM, and ANN in order to increase explainability of the given black box model. The algorithms were trained on data from the National Highway Traffic Safety Administration (NHTSA) and the Comprehensive State-by-State Reporting (CSSR) system and the Bayesian optimization technique was used to obtain optimal hyperparameters. XGBoost outperformed the other black box machine learning models in identifying severe traffic accidents with Random Forest close behind. However, performance in identifying traffic accidents did not correspond to explainability as determined by each XAI model, which varied depending on the black box model and the XAI model used. The paper provides in-depth background information about the studies that have been done in the field and the mathematical underpinnings of the techniques used. Credibility is enhanced by its recency, its peer-reviewed status, and its wealth of background information which provide solid reasoning

for steps taking in the research.

[11] The study utilizes the decision tree classifier (DT), gradient boosting classifier (GBC), and random forest classifier (RFC) machine leraning models to predict traffic flow on Ikorodu Road in Lagos State, Nigeria. As a rapidly urbanizing and developing city, Lagos lacks sufficient infrastructure to meet traffic demand, as well as a lack of traditional data sources typically used in machine learning projects. Traffic sensors and cameras were used to gather data, and descriptive statistics were used alongside machine learning models, which classified the traffic into "High," "Medium," and "Low." GBC had the highest cross validation score while RFC had the highest accuracy score. The paper concludes with recommendations on some limitations of the study and how policymakers can implement its findings. The methods and purpose of this paper aligns closely with other papers on the subject. As for its creditibility, it was published recently in 2025 and was cited once by another paper.

[16] The study compared four machine learning algorithms used for traffic prediction: K-Nearest Neighbors (KNN), Naïve Bayes, Decision Tree, and Random Forest. The algorithms received data on vehicle count in a given location and had to classify it as high traffic, medium traffic, no traffic, and traffic jam. Random Forest performed the best, with K-Nearest Neighbors close behind but slightly hampered by its high computational resource requirement. Naïve Bayers performed the worst in terms of precision and recall. The article is published in IEEE and was published recently (2025), so its crediblity and relevance is relatively high, however, it only has 5 full text views, which may detract from its credibility.

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